

Week 7: Operational Semantics & Hoare Logic

1 Operational Semantics

This section is about the small-step operational semantics of While programs as given by the relation $\rightarrow \subseteq \mathcal{C} \times \mathcal{C}$ where the configurations are either a statement and a state or a state or a state $\mathcal{C} = (S \times \text{State}) \cup \text{State}$, which is defined by these following rules:

$$\begin{array}{c}
 \frac{}{\langle \text{skip}, \sigma \rangle \rightarrow \sigma} \qquad \frac{}{\langle x \leftarrow e, \sigma \rangle \rightarrow \sigma[x \mapsto \llbracket e \rrbracket_{\mathcal{A}}(\sigma)]} \\
 \\
 \frac{\langle S_1, \sigma_1 \rangle \rightarrow \sigma_2}{\langle S_1; S_2, \sigma_1 \rangle \rightarrow \langle S_2, \sigma_2 \rangle} \qquad \frac{\langle S_1, \sigma_1 \rangle \rightarrow \langle S'_1, \sigma_2 \rangle}{\langle S_1; S_2, \sigma_1 \rangle \rightarrow \langle S'_1; S_2, \sigma_2 \rangle} \\
 \\
 \frac{}{\langle \text{if } e \text{ then } S_1 \text{ else } S_2, \sigma \rangle \rightarrow \langle S_1, \sigma \rangle} \llbracket e \rrbracket_{\mathcal{B}}(\sigma) = \top \\
 \\
 \frac{}{\langle \text{if } e \text{ then } S_1 \text{ else } S_2, \sigma \rangle \rightarrow \langle S_2, \sigma \rangle} \llbracket e \rrbracket_{\mathcal{B}}(\sigma) = \perp \\
 \\
 \frac{}{\langle \text{while } e \text{ do } S, \sigma \rangle \rightarrow \sigma} \llbracket e \rrbracket_{\mathcal{B}}(\sigma) = \perp \\
 \\
 \frac{}{\langle \text{while } e \text{ do } S, \sigma \rangle \rightarrow \langle S; \text{while } e \text{ do } S, \sigma \rangle} \llbracket e \rrbracket_{\mathcal{B}}(\sigma) = \top
 \end{array}$$

Figure 1: Operational semantics for While.

- * 1. Calculate the terminal state and write down a trace for the statement $x \leftarrow 1; \{x \leftarrow 2; x \leftarrow 3\}$ with the initial state $[]$ using the rules in Figure 1. Remember, variables are assigned 0 by default.

Solution

The terminal state is $[x \mapsto 3]$ and the corresponding trace is:

$$\begin{aligned}
 &\langle x \leftarrow 1; \{x \leftarrow 2; x \leftarrow 3\}, [] \rangle \\
 &\rightarrow \langle x \leftarrow 2; x \leftarrow 3, [x \mapsto 1] \rangle \\
 &\rightarrow \langle x \leftarrow 3, [x \mapsto 2] \rangle \\
 &\rightarrow [x \mapsto 3]
 \end{aligned}$$

- * 2. Calculate the terminal state and write down a trace for the statement $\{x \leftarrow 1; x \leftarrow x * 2\}; x \leftarrow x + y$ with the initial state $[x \mapsto 2, y \mapsto 2]$ using the rules in Figure 1.

Solution

The terminal state is $[x \mapsto 4, y \mapsto 2]$ and the corresponding trace is:

$$\begin{aligned} & \langle \{x \leftarrow 1; x \leftarrow x * 2\}; x \leftarrow x + y, [x \mapsto 2, y \mapsto 2] \rangle \\ & \rightarrow \langle x \leftarrow x * 2; x \leftarrow x + y, [x \mapsto 1, y \mapsto 2] \rangle \\ & \rightarrow \langle x \leftarrow x + y, [x \mapsto 2, y \mapsto 2] \rangle \\ & \rightarrow [x \mapsto 4, y \mapsto 2] \end{aligned}$$

- * 3. Find a state σ such that $\langle x \leftarrow 1; y \leftarrow x * 2, [] \rangle \rightarrow^* \sigma$. You should provide the corresponding trace.

Solution

The state $[x \mapsto 1, y \mapsto 2]$ satisfies the requirement with the associated trace:

$$\begin{aligned} & \langle x \leftarrow 1; y \leftarrow x * 2, [] \rangle \\ & \rightarrow \langle y \leftarrow x * 2, [x \mapsto 1] \rangle \\ & \rightarrow [x \mapsto 1, y \mapsto 2] \end{aligned}$$

- * 4. Compute the final state for the program if $x \leq y$ then $x \leftarrow y$ else $y \leftarrow x$ when executed in each of the following states:

- $[]$
- $[x \mapsto 2, y \mapsto 3]$
- $[x \mapsto 4, y \mapsto 2]$

Solution

- $[]$
- $[x \mapsto 3, y \mapsto 3]$
- $[x \mapsto 4, y \mapsto 4]$

- * 5. Find a state $\sigma \in \text{State}$ such that $\text{while } !(x \leq -1) \text{ do } x \leftarrow x + d, [d \mapsto -1] \rightarrow^* \sigma$. You should provide the corresponding trace.

Solution

The state $[d \mapsto -1, x \mapsto -1]$ will satisfy the requirements. The associated trace is:

$$\begin{aligned} & \langle \text{while } !(x \leq -1) \text{ do } x \leftarrow x + d, [d \mapsto -1] \rangle \\ & \rightarrow \langle x \leftarrow x + d; \text{while } !(x \leq -1) \text{ do } x \leftarrow x + d, [d \mapsto -1] \rangle \\ & \rightarrow \langle \text{while } !(x \leq -1) \text{ do } x \leftarrow x + d, [d \mapsto -1, x \mapsto -1] \rangle \\ & \rightarrow [d \mapsto -1, x \mapsto -1] \end{aligned}$$

- * 6. Find a state $\sigma \in \text{State}$ such that $\langle x \leftarrow 2; y \leftarrow x * y, \sigma \rangle \rightarrow^* [x \mapsto 2, y \mapsto 4]$. You should provide the corresponding trace.

Solution

The state $[y \mapsto 2]$ will satisfy the requirements. The associated trace is:

$$\begin{aligned} & \langle x \leftarrow 2; y \leftarrow x * y, [y \mapsto 2] \rangle \\ & \rightarrow \langle y \leftarrow x * y, [x \mapsto 2, y \mapsto 2] \rangle \\ & \rightarrow [x \mapsto 2, y \mapsto 4] \end{aligned}$$

- ** 7. Suppose $e \in \mathcal{B}$ is a Boolean expression that is semantically equivalent to false. Prove that $\langle \text{while } e \text{ do } S, \sigma \rangle \rightarrow \sigma$ for any state $\sigma \in \text{State}$.

Solution

As $e \in \mathcal{B}$ is semantically equivalent to false. We have that $\llbracket e \rrbracket_{\mathcal{B}}(\sigma) = \perp$ for any state $\sigma \in \text{State}$. Therefore, we have that $\langle \text{while } e \text{ do } S, \sigma \rangle \rightarrow \sigma$ for any state $\sigma \in \text{State}$ as required.

- ** 8. Suppose $S_1, S_2 \in \mathcal{S}$ are two statements such that $\langle S_1, \sigma \rangle \rightarrow^* \sigma'$ and $\langle S_2, \sigma \rangle \rightarrow^* \sigma'$ for some states $\sigma, \sigma' \in \text{State}$. Prove that $\langle \text{if } e \text{ then } S_1 \text{ else } S_2, \sigma \rangle \rightarrow^* \sigma'$ for any Boolean expression $e \in \mathcal{B}$.

Solution

Let $e \in \mathcal{B}$ be some Boolean expression and $\sigma, \sigma' \in \text{State}$ be two states. Let us consider two cases:

- Suppose $\llbracket e \rrbracket_{\mathcal{B}}(\sigma)$ is true. Then we have the following trace:

$$\begin{aligned} & \langle \text{if } e \text{ then } S_1 \text{ else } S_2, \sigma \rangle \\ & \rightarrow \langle S_1, \sigma \rangle \\ & \rightarrow^* \sigma' \end{aligned}$$

- Suppose, otherwise, that $\llbracket e \rrbracket_{\mathcal{B}}(\sigma)$ is false. Then equally we have the following trace of the form:

$$\begin{aligned} & \langle \text{if } e \text{ then } S_1 \text{ else } S_2, \sigma \rangle \\ & \rightarrow \langle S_2, \sigma \rangle \\ & \rightarrow^* \sigma' \end{aligned}$$

- ** 9. Suppose we introduce a new language construct “do S while e ” where $S \in \mathcal{S}$ is a statement and $e \in \mathcal{B}$ is a Boolean expression. The operational semantics for this construct is given by the following inference rules:

$$\frac{}{\langle \text{do } S \text{ while } e, \sigma \rangle \rightarrow \langle S; \text{if } e \text{ then do } S \text{ while } e \text{ else skip}, \sigma \rangle}$$

- Find a state $\sigma \in \text{State}$ such that $\langle \text{do } x \leftarrow x + 1 \text{ while } x \leq 1, [] \rangle \rightarrow^* \sigma$ and give the associated trace.
- For a given statement $S \in \mathcal{S}$ and a Boolean expression $e \in \mathcal{B}$ find a While program that is equivalent to the program do S while e but does not use the new construct. That is, find a statement $S' \in \mathcal{S}$ such that:

$$\langle S', \sigma \rangle \rightarrow^* \sigma' \Leftrightarrow \langle \text{do } S \text{ while } e, \sigma \rangle \rightarrow^* \sigma'$$

You do not need to prove that your answer is correct but should provide a trace for

$\langle S', [] \rangle \rightarrow^* \sigma$ where S is given to be the statement $x \leftarrow x + 1$, where e is given to be the expression $x \leq 1$, and where σ is the state from part (a).

Solution

(a) The state is $[x \mapsto 2]$ and the trace is:

$\langle \text{do } x \leftarrow x + 1 \text{ while } x \leq 1, [] \rangle$
 $\rightarrow \langle x \leftarrow x + 1; \text{if } x \leq 1 \text{ then do } x \leftarrow x + 1 \text{ while } x \leq 1 \text{ else skip}, [] \rangle$
 $\rightarrow \langle \text{if } x \leq 1 \text{ then do } x \leftarrow x + 1 \text{ while } x \leq 1 \text{ else skip}, [x \mapsto 1] \rangle$
 $\rightarrow \langle \text{do } x \leftarrow x + 1 \text{ while } x \leq 1, [x \mapsto 1] \rangle$
 $\rightarrow \langle x \leftarrow x + 1; \text{if } x \leq 1 \text{ then do } x \leftarrow x + 1 \text{ while } x \leq 1 \text{ else skip}, [x \mapsto 1] \rangle$
 $\rightarrow \langle \text{if } x \leq 1 \text{ then do } x \leftarrow x + 1 \text{ while } x \leq 1 \text{ else skip}, [x \mapsto 2] \rangle$
 $\rightarrow \langle \text{skip}, [x \mapsto 2] \rangle$
 $\rightarrow [x \mapsto 2]$

(b) The statement S ; while e do S is equivalent.

$\langle x \leftarrow x + 1; \text{while } x \leq 1 \text{ do } x \leftarrow x + 1, [] \rangle$
 $\rightarrow \langle \text{while } x \leq 1 \text{ do } x \leftarrow x + 1, [x \mapsto 1] \rangle$
 $\rightarrow \langle x \leftarrow x + 1; \text{while } x \leq 1 \text{ do } x \leftarrow x + 1, [x \mapsto 1] \rangle$
 $\rightarrow \langle \text{while } x \leq 1 \text{ do } x \leftarrow x + 1, [x \mapsto 2] \rangle$
 $\rightarrow [x \mapsto 2]$

** 10. Suppose we introduce a new language construct for x do S where $S \in \mathcal{S}$ is a statement and $x \in \text{Var}$ is a variable. The operational semantics for this construct is given by the following inference rules:

$$\frac{}{\langle \text{for } x \text{ do } S, \sigma \rangle \rightarrow \sigma} \sigma(x) \leq 0 \quad \frac{}{\langle \text{for } x \text{ do } S, \sigma_1 \rangle \rightarrow \langle S; \text{for } x \text{ do } S, \sigma[x \mapsto \sigma(x) - 1] \rangle} \sigma(x) > 0$$

- (a) Find a state $\sigma \in \text{State}$ such that $\langle \text{for } x \text{ do } y \leftarrow y + x; x \leftarrow x - 2, [x \mapsto 3] \rangle \rightarrow^* \sigma$ and give the associated trace.
- (b) For a given statement $S \in \mathcal{S}$ and a variable $x \in \text{Var}$ find a While program that is equivalent to the program for x do S but does not use the new construct. That is, find a statement $S' \in \mathcal{S}$ such that:

$$\langle S', \sigma \rangle \rightarrow^* \sigma' \Leftrightarrow \langle \text{for } x \text{ do } S, \sigma \rangle \rightarrow^* \sigma'$$

You do not need to prove that your answer is correct but should provide a trace for $\langle S', [x \mapsto 3] \rangle \rightarrow^* \sigma$ where S is given to be the statement $y \leftarrow y + x; x \leftarrow x - 2$ and σ is the state from part (a).

Solution

(a)

$$\begin{aligned} & \langle \text{for } x \text{ do } y \leftarrow y + x; x \leftarrow x - 2, [x \mapsto 3] \rangle \\ & \rightarrow \langle y \leftarrow y + x; x \leftarrow x - 2; \text{for } x \text{ do } y \leftarrow y + x; x \leftarrow x - 2, [x \mapsto 2] \rangle \\ & \rightarrow \langle x \leftarrow x - 2; \text{for } x \text{ do } y \leftarrow y + x; x \leftarrow x - 2, [x \mapsto 2, y \mapsto 2] \rangle \\ & \rightarrow \langle \text{for } x \text{ do } y \leftarrow y + x; x \leftarrow x - 2, [x \mapsto 0, y \mapsto 2] \rangle \\ & \rightarrow [x \mapsto 0, y \mapsto 2] \end{aligned}$$

(b) The statement $\text{while } !(x \leq 0) \text{ do } \{x \leftarrow x - 1; S\}$ is equivalent.

$$\begin{aligned} & \langle \text{while } !(x \leq 0) \text{ do } x \leftarrow x - 1; y \leftarrow y + x; x \leftarrow x - 2, [x \mapsto 3] \rangle \\ & \rightarrow \langle x \leftarrow x - 1; y \leftarrow y + x; x \leftarrow x - 2; \text{while } !(x \leq 0) \text{ do } x \leftarrow x - 1; y \leftarrow y + x; x \leftarrow x - 2, [x \mapsto 3] \rangle \\ & \rightarrow^* \langle \text{while } !(x \leq 0) \text{ do } x \leftarrow x - 1; y \leftarrow y + x; x \leftarrow x - 2, [x \mapsto 0, y \mapsto 2] \rangle \\ & \rightarrow [x \mapsto 0, y \mapsto 2] \end{aligned}$$

- *** 11. Show that, if $y \notin \text{FV}(e_1)$ and $x \notin \text{FV}(e_2)$, then the statements $x \leftarrow e_1; y \leftarrow e_2$ and $y \leftarrow e_2; x \leftarrow e_1$ are equivalent in the sense that $\langle x \leftarrow e_1; y \leftarrow e_2, \sigma \rangle \rightarrow^* \sigma'$ if and only if $\langle y \leftarrow e_2; x \leftarrow e_1, \sigma \rangle \rightarrow^* \sigma'$. You may use results from the previous worksheet.

Solution

This is most straightforward proven from the fact that $\llbracket e_1 \rrbracket_{\mathcal{A}}(\sigma) = \llbracket e_1 \rrbracket_{\mathcal{A}}(\sigma')$ if $\sigma(x) = \sigma'(x)$ for all $x \in \text{FV}(e_1)$, and likewise for e_2 .

Note that $\langle x \leftarrow e_1; y \leftarrow e_2, \sigma_0 \rangle \rightarrow^* \sigma_2$ just if $\sigma_1 = \sigma_0[x \mapsto \llbracket e_1 \rrbracket_{\mathcal{A}}(\sigma_0)]$ and $\sigma_2 = \sigma_1[y \mapsto \llbracket e_2 \rrbracket_{\mathcal{A}}(\sigma_1)]$. And, likewise $\langle y \leftarrow e_2; x \leftarrow e_1, \sigma_0 \rangle \rightarrow^* \sigma'_2$ just if $\sigma'_1 = \sigma_0[y \mapsto \llbracket e_2 \rrbracket_{\mathcal{A}}(\sigma_0)]$ and $\sigma'_2 = \sigma'_1[x \mapsto \llbracket e_1 \rrbracket_{\mathcal{A}}(\sigma'_1)]$.

To show that σ_2 is equal to σ'_2 , we must show that $\llbracket e_1 \rrbracket_{\mathcal{A}}(\sigma_0) = \llbracket e_1 \rrbracket_{\mathcal{A}}(\sigma'_1)$ and that $\llbracket e_2 \rrbracket_{\mathcal{A}}(\sigma_1) = \llbracket e_2 \rrbracket_{\mathcal{A}}(\sigma'_2)$. Using the previous result, and the fact that $\sigma_0(z) = \sigma'_1(z)$ for all $z \neq y$, we have that $\llbracket e_1 \rrbracket_{\mathcal{A}}(\sigma_0) = \llbracket e_1 \rrbracket_{\mathcal{A}}(\sigma'_1)$ as $y \notin \text{FV}(e_1)$. And likewise $\llbracket e_2 \rrbracket_{\mathcal{A}}(\sigma_1) = \llbracket e_2 \rrbracket_{\mathcal{A}}(\sigma'_2)$ follows from $x \notin \text{FV}(e_2)$. Therefore, $\sigma_2 = \sigma'_2$ as required.

- *** 12. The set of variables *modified* by a statement is defined by recursion as follows:

$$\begin{aligned} \text{mod}(\text{skip}) &= \emptyset \\ \text{mod}(x \leftarrow e) &= \{x\} \\ \text{mod}(S_1; S_2) &= \text{mod}(S_1) \cup \text{mod}(S_2) \\ \text{mod}(\text{if } e \text{ then } S_1 \text{ else } S_2) &= \text{mod}(S_1) \cup \text{mod}(S_2) \\ \text{mod}(\text{while } e \text{ do } S) &= \text{mod}(S) \end{aligned}$$

- (a) Prove that if $\langle S_1, \sigma_1 \rangle \rightarrow \sigma_2$ then $\sigma_1(x) = \sigma_2(x)$ for all $x \notin \text{mod}(S_1)$.
- (b) Prove that if $\langle S_1, \sigma_1 \rangle \rightarrow \langle S_1, \sigma_2 \rangle$ then $\text{mod}(S_2) \subseteq \text{mod}(S_1)$ and $\sigma_1(x) = \sigma_2(x)$ for all $x \notin \text{mod}(S_1)$. You may use the previous result.
- (c) Prove that if $\langle S_1, \sigma_1 \rangle \rightarrow^* \sigma_2$ then $\sigma_1(x) = \sigma_2(x)$ for all $x \notin \text{mod}(S_1)$. You may use the previous results, and the fact that $\gamma_1 \rightarrow^* \gamma_2$ if, and only if, $\gamma_1 \rightarrow^n \gamma_2$ for some $n \geq 0$.

Solution

- (a) First, we shall prove that $\langle S_1, \sigma_1 \rangle \rightarrow \sigma_2$ then $\sigma_1(x) = \sigma_2(x)$ for all $x \notin \text{mod}(S_1)$. There are only three cases for this:
- If $\langle \text{skip}, \sigma \rangle \rightarrow \sigma$ then we immediately have that $\sigma(x) = \sigma(x)$ for all $x \notin \text{mod}(\text{skip})$ as required.
 - If $\langle x \leftarrow e, \sigma \rangle \rightarrow \sigma[x \mapsto \llbracket e \rrbracket_{\mathcal{A}}(\sigma)]$ then we have that $\sigma(y) = \sigma[x \mapsto \llbracket e \rrbracket_{\mathcal{A}}(\sigma)](y)$ for all $y \notin \text{mod}(x \leftarrow e)$ as $x \in \text{mod}(x \leftarrow e)$.
 - If $\langle \text{while } e \text{ do } S, \sigma \rangle \rightarrow \sigma$ when $\llbracket e \rrbracket_{\mathcal{B}}(\sigma) = \perp$, then again we immediately have that $\sigma(x) = \sigma(x)$ for all $x \notin \text{mod}(\text{skip})$ as required.
- (b) To complete the second part of this proof, you need to use induction over the statement. However, it is only relevant in the sequence case(s) — the induction hypothesis is not important for any other case.
- We shall prove that if $\langle S_1, \sigma_1 \rangle \rightarrow \langle S_1, \sigma_2 \rangle$ then $\text{mod}(S_2) \subseteq \text{mod}(S_1)$ and $\sigma_1(x) = \sigma_2(x)$ for all $x \notin \text{mod}(S_1)$ by structural induction over S_1 .
- In the case of skip or an assignment, there is nothing to prove as it steps to a terminal configuration.
 - For the statement $S_1; S_2$, we have $\langle S_1; S_2, \sigma_1 \rangle \rightarrow \langle S'_1; S_2, \sigma_2 \rangle$ when $\langle S_1, \sigma_1 \rangle \rightarrow \langle S'_1, \sigma_2 \rangle$. By induction, we have that $\text{mod}(S'_1) \subseteq \text{mod}(S_1)$ and $\sigma_1(x) = \sigma_2(x)$ for all $x \notin \text{mod}(S_1)$. It then follows that $\text{mod}(S'_1; S_2) \subseteq \text{mod}(S_1; S_2)$ and $\sigma_1(x) = \sigma_2(x)$ for all $x \notin \text{mod}(S_1; S_2)$ as $\text{mod}(S_1) \subseteq \text{mod}(S_1; S_2)$.
If, on the other hand, we have $\langle S_1; S_2, \sigma_1 \rangle \rightarrow \langle S_2, \sigma_2 \rangle$ when $\langle S_1, \sigma_1 \rangle \rightarrow \sigma_2$, the by the previous result $\sigma_1(x) = \sigma_2(x)$ for all $x \notin \text{mod}(S_1)$ and thus by extension $\sigma_1(x) = \sigma_2(x)$ for all $x \notin \text{mod}(S_1; S_2)$ as $\text{mod}(S_1) \subseteq \text{mod}(S_1; S_2)$.
 - For the if e then S_1 else S_2 , we either have $\langle \text{if } e \text{ then } S_1 \text{ else } S_2, \sigma \rangle \rightarrow \langle S_1, \sigma \rangle$ or $\langle \text{if } e \text{ then } S_1 \text{ else } S_2, \sigma_1 \rangle \rightarrow \langle S_2, \sigma \rangle$. In either case, we have that $\text{mod}(S_1), S_2 \subseteq \text{mod}(\text{if } e \text{ then } S_1 \text{ else } S_2)$. Additionally, $\sigma(x) = \sigma(x)$ for all $x \notin \text{mod}(\text{if } e \text{ then } S_1 \text{ else } S_2)$.
 - The case of while is analogous to the former.
- (c) The final part of this exercise is to prove that $\langle S_1, \sigma_1 \rangle \rightarrow^n \sigma_2$ then $\sigma_1(x) = \sigma_2(x)$ for all $x \notin \text{mod}(S_1)$. This is done by induction over n , i.e. the length of the trace.
- The base case is absurd as to reach a termination configuration from a non-termination configuration requires at least one step.
 - Suppose that $\langle S_1, \sigma_1 \rangle \rightarrow \gamma$ and $\gamma \rightarrow^n \sigma_2$. There are two cases to consider:
 - γ is the terminal configuration σ_2 . In which case, by the previous results, $\sigma_1(x) = \sigma_2(x)$ for all $x \notin \text{mod}(S_1)$ as required.
 - γ is another non-terminal configuration $\langle S_2, \sigma_3 \rangle$. In which case, by the previous results, $\sigma_1(x) = \sigma_3(x)$ for all $x \notin \text{mod}(S_1)$ and $\text{mod}(S_2) \subseteq \text{mod}(S_1)$. By the induction hypothesis, $\sigma_3(x) = \sigma_2(x)$ for all $x \notin \text{mod}(S_2)$, and so $\sigma_3(x) = \sigma_2(x)$ for all $x \notin \text{mod}(S_1)$ as $\text{mod}(S_2) \subseteq \text{mod}(S_1)$. It follows that $\sigma_1(x) = \sigma_2(x)$ for all $x \notin \text{mod}(S_1)$ as required.

Hoare Logic

- * 13. Consider the following *invalid* Hoare triple: $\{x \leq y\} \ y \leftarrow y * 2 \ \{x \leq y\}$. Find an initial state $\sigma \in \text{State}$ and a trace from the configuration $\langle y \leftarrow y * 2, \sigma \rangle$ that contradicts this triple.

Solution

Any state with $2y < x \leq y$ suffices, in particular both x and y must be negative.

$$\langle y \leftarrow y * 2, [x \mapsto -3, y \mapsto -2] \rangle \\ \rightarrow [x \mapsto -3, y \mapsto -4]$$

- * 14. Find a statement S such that $\{\text{true}\} S \{z \leq x \ \&\& \ z \leq y\}$.

Solution

There are multiple possible solutions. One is a program that calculates the minimum:

$$\text{if } x \leq y \text{ then } z \leftarrow x \text{ else } z \leftarrow y$$

- ** 15. For each of the following statements, compute the strongest post-condition from the pre-condition $(\exists q. x = q * y) \ \&\& \ y \geq 0$:

(a) $x \leftarrow x * x$

(b) $\text{if } y \leq x \text{ then } x \leftarrow x - y \text{ else } y \leftarrow y - x$

Solution

(a) Any variation of: $\exists x'. (\exists q. x' = q * y) \ \&\& \ x = x' * x' \ \&\& \ y \geq 0$. More simply, $y \geq 0 \ \&\& \ \exists q. x = (q * y) * (q * y)$.

(b) Any variation of:

$$(\exists x'. y \leq x' \ \&\& \ (\exists q. x' = q * y) \ \&\& \ y \geq 0 \ \&\& \ x = x' - y) \\ \parallel (\exists y'. y' > x \ \&\& \ (\exists q. x = q * y') \ \&\& \ y' \geq 0 \ \&\& \ y = y' - x)$$

More simply, $(0 \leq y \ \&\& \ 0 \leq x \ \&\& \ (\exists q. x = q * y)) \parallel (y > 0 \ \&\& \ y + x \geq 0 \ \&\& \ (\exists q. x = q * (y + x)))$.

- ** 16. Using your answers to the previous question, for each of the following Hoare triples determine whether they are valid or not. You should justify your answer.

(a) $\{(\exists q. x = q * y) \ \&\& \ y \geq 0\} x \leftarrow x * x \{(\exists q. x = q * y)\}$

(b) $\{(\exists q. x = q * y) \ \&\& \ y \geq 0\} x \leftarrow x * x \{x \geq y\}$

(c) $\{(\exists q. x = q * y) \ \&\& \ y \geq 0\} \text{if } y \leq x \text{ then } x \leftarrow x - y \text{ else } y \leftarrow y - x \{(\exists q. x = q * y) \ \&\& \ y \geq 0\}$

Solution

(a) This is valid as the strongest post-condition $\exists x'. (\exists q. x' = q * y) \ \&\& \ x = x' * x' \ \&\& \ y \geq 0$ implies $\exists q. x = q * y$. In particular, suppose $x' = q * y$ and $x = x' * x'$. Then $x = (q * y) * (q * y) = (q^2 * y) * y$ as required.

(b) This is not valid as the strongest post-condition $\exists x'. (\exists q. x' = q * y) \ \&\& \ x = x' * x' \ \&\& \ y \geq 0$ does not imply $x \geq y$. In particular, suppose $x' = q * y$ and $x = x' * x'$. However, we could have $q = 0$ and $y = 1$, then $x = 0$ and we do not have $x \geq y$. If we additionally knew that $q \neq 0$, then it would be valid.

(c) This is not valid as the strongest post-condition:

$$\begin{aligned} & (\exists x'. y \leq x' \ \&\& \ (\exists q. x' = q * y) \ \&\& \ y \geq 0 \ \&\& \ x = x' - y) \\ & \parallel (\exists y'. y' > x \ \&\& \ (\exists q. x = q * y') \ \&\& \ y' \geq 0 \ \&\& \ y = y' - x) \end{aligned}$$

does not imply $(\exists q. x = q * y) \ \&\& \ y \geq 0$.

In particular, out of the post-condition of both branches $\exists x'. y \leq x' \ \&\& \ (\exists q. x' = q * y) \ \&\& \ y \geq 0 \ \&\& \ x = x' - y$ and $\exists y'. y' > x \ \&\& \ (\exists q. x = q * y') \ \&\& \ y' \geq 0 \ \&\& \ y = y' - x$, only the former implies $(\exists q. x = q * y) \ \&\& \ y \geq 0$:

- Suppose $y \geq x'$ and $x' = q * y$ and $y \geq 0$ where $x = x' - y$. Then $x = (q - 1) * y$ and thus $(\exists q. x = q * y) \ \&\& \ y \geq 0$. As $y \geq 0$ by assumption, the desired conclusion holds.
- Now suppose that $y' > x$ and $x = q * y'$ and $y' \geq 0$ where $y = y' - x$. We have that $y + x > x$ and so $y \geq 0$. However, we cannot conclude that $x = q * y$ for some q given that $x = q * (y + x)$ for some q . For instance, we could have $x = -1$, $y = 2$, and $q = -1$. This implies that $y' = 1$ (i.e. the old value of y) and so the triple is invalidated by the initial state $[x \mapsto -1, y \mapsto 1]$.

** 17. Suppose $S_1, S_2, S_3 \in \mathcal{S}$ are statements satisfying the following Hoare triples:

- $\{x \leq y\} S_1 \{y \leq x \ \&\& \ x \leq 0\}$
- $\{\forall z. x * z = 0\} S_2 \{y = z\}$
- $\{x > y \ \&\& \ z = x\} S_3 \{y \leq x\}$

Then which of the following triples can be derived? You should briefly justify your answer.

- (a) $\{x \leq y \ \&\& \ x = 0\} S_1; S_2 \{y = z\}$
- (b) $\{z = x\}$ if $x \leq y$ then S_1 else $S_2 \{y \leq x\}$
- (c) $\{x \leq y\} S_1$; if $y = 0$ then S_2 else $y \leftarrow z \{y = z\}$

Solution

- (a) This triple cannot be derived as we only know that $x = 0$ prior to the execution of S_1 . Therefore, we cannot conclude that $\forall z. x * z = 0$ prior to the execution of S_2 .
- (b) For the original question $\{z = x\}$ if $x \leq y$ then S_1 else $S_2 \{y \leq x\}$, the answer is false — it is not a valid triple as the second branch does not give us that $y \leq x$. However, this was a typo... The question was meant to be: $\{z = x\}$ if $x \leq y$ then S_1 else $S_3 \{y \leq x\}$. In which case the triple can be derived as we have that: $\{x \leq y \ \&\& \ z = x\} S_1 \{y \leq x\}$ via weakening for the first branch.
- (c) This triple can be derived. First, $\{\forall z. x * z = 0\} S_2 \{y = z\}$ can be weakened to $\{x = 0\} S_2 \{y = z\}$. Therefore, $\{y \leq x \ \&\& \ x \leq 0\}$ if $y = 0$ then S_2 else $y \leftarrow z \{y = z\}$. So, by

sequencing $\{x \leq y\} S_1 \{y \leq x \ \&\& \ x \leq 0\}$ and $\{y \leq x \ \&\& \ x \leq 0\}$ if $y = 0$ then S_2 else $z \leftarrow y \{y = z\}$. We have $\{x \leq y\} S_1$; if $y = 0$ then S_2 else $y \leftarrow z \{y = z\}$ as required.

- *** 18. The rule of consequence allows us to weaken a Hoare triple for more general context. However, it is also useful to be able to adapt Hoare triples to contexts with other unrelated variables. This can be done through the *constancy rule*:

$$\frac{\{P\} S \{Q\}}{\{P \ \&\& \ R\} S \{Q \ \&\& \ R\}} \text{FV}(R) \cap \text{mod}(S) = \emptyset$$

Where $\text{FV}(P)$ for a predicate $P \subseteq \text{State}$ is defined as the set of variables $x \in \text{Var}$ such that $\sigma \in P$ but $\sigma[x \mapsto n] \notin P$ for some state $\sigma \in \text{State}$ and $n \in \mathbb{Z}$; intuitively, those variables whose value effect whether a state is in the set of not and $\text{mod}(S)$ is the set of variables appearing anywhere in the program.

- (a) Suppose that S_1 and S_2 are two statement such that $\{P_1\} S_1 \{Q_1\}$ and $\{P_2\} S_2 \{Q_2\}$ where:
- $\text{FV}(P_1), \text{FV}(Q_1) \subseteq \text{FV}(S_1)$;
 - $\text{FV}(P_2), \text{FV}(Q_2) \subseteq \text{FV}(S_2)$;
 - And $\text{FV}(S_1) \cap \text{FV}(S_2) = \emptyset$.

Justify how $\{P_1 \wedge P_2\} S_1; S_2 \{Q_1 \wedge Q_2\}$ can be derived from the constancy rule.

- (b) Suppose that $P \subseteq \text{State}$ is some set of states. Prove that, if $\sigma \in P$ and $\sigma(x) = \sigma'(x)$ for all $x \in \text{FV}(P)$ and $\#\{x \in \text{Var} \mid \sigma(x) \neq \sigma'(x)\}$ is finite, then $\sigma' \in P$. You may wish to prove it by induction on $\#\{x \in \text{Var} \mid \sigma(x) \neq \sigma'(x)\}$.
- (c) Prove that the rule of constancy holds that is:

$$\{P\} S \{Q\} \Rightarrow \{P \ \&\& \ R\} S \{Q \ \&\& \ R\}$$

for any statement S , and predicates $P, Q, R \subseteq \text{State}$ such that $\text{FV}(R) \cap \text{mod}(S) = \emptyset$. You may wish to use the result from Question 12.

Solution

- (a) By the constancy rule, we have that $\{P_1 \ \&\& \ P_2\} S_1 \{Q_1 \ \&\& \ P_2\}$ as $\text{FV}(P_2) \subseteq \text{FV}(S_2)$ which is disjoint from $\text{FV}(S_1)$. Likewise, we have that $\{Q_1 \ \&\& \ P_2\} S_2 \{Q_1 \ \&\& \ Q_2\}$. Therefore, $\{P_1 \ \&\& \ P_2\} S_1; S_2 \{Q_1 \ \&\& \ Q_2\}$ as required.
- (b) Let us write $\text{dis}(\sigma, \sigma')$ for $\#\{x \in \text{Var} \mid \sigma(x) \neq \sigma'(x)\}$.
Let $P \subseteq \text{State}$ be a set of states. We shall prove by induction on n that if:

$$\phi(n) : \forall \sigma, \sigma' \in \text{State}. \text{dis}(\sigma, \sigma') = n \text{ and } \forall x \in \text{FV}(P). \sigma(x) = \sigma'(x) \text{ then } \sigma' \in P$$

- In the base case, we have that there are no variables such that $\sigma(x) \neq \sigma'(x)$. Therefore, $\sigma = \sigma'$ and $\sigma' \in P$ by assumption.
- Now suppose $\phi(n)$ holds for some n . Consider two states $\sigma, \sigma' \in \text{State}$ such that $\text{dis}(\sigma, \sigma') = n + 1$ and $\forall x \in \text{FV}(P). \sigma(x) = \sigma'(x)$. Let y be some variable such that $\sigma(y) \neq \sigma'(y)$ (which must exist as they disagree on $n + 1$ variables). Note that $y \notin \text{FV}(P)$, else we'd contradict the assumption that $\forall x \in \text{FV}(P). \sigma(x) = \sigma'(x)$.

Now consider the state $\sigma[y \mapsto \sigma'(y)]$. We have that $\text{dis}(\sigma[y \mapsto \sigma'(y)], \sigma') = n$ and $\sigma[y \mapsto \sigma'(y)](x) = \sigma'(x)$ for all $x \in \text{FV}(P)$. Suppose that $\sigma[y \mapsto \sigma'(y)] \notin P$. Then, by definition, $y \in \text{FV}(P)$, which is contradictory. Thus, $\sigma[y \mapsto \sigma'(y)] \in P$. By $\phi(n)$, therefore, $\sigma' \in P$.

- (c) Question 12 tells us that if $\langle S, \sigma \rangle \rightarrow^* \sigma'$, then $\sigma(x) = \sigma'(x)$ for all $x \notin \text{mod}(S)$. Now suppose $\sigma \in P \ \&\& \ R$, i.e. it meets the pre-condition. From the rule's premise, we have that $\sigma' \in Q$. To show that $\sigma' \in R$, we must consider the fact that $\text{mod}(S) \cap \text{FV}(R) = \emptyset$. Therefore, $\sigma(x) = \sigma'(x)$ for all $x \in \text{FV}(R)$. As σ and σ' can only differ on variables in S , by the previous exercise, $\sigma' \in R$. Therefore, $\sigma' \in P \ \&\& \ R$ as required.